



Neural Networks Foundations

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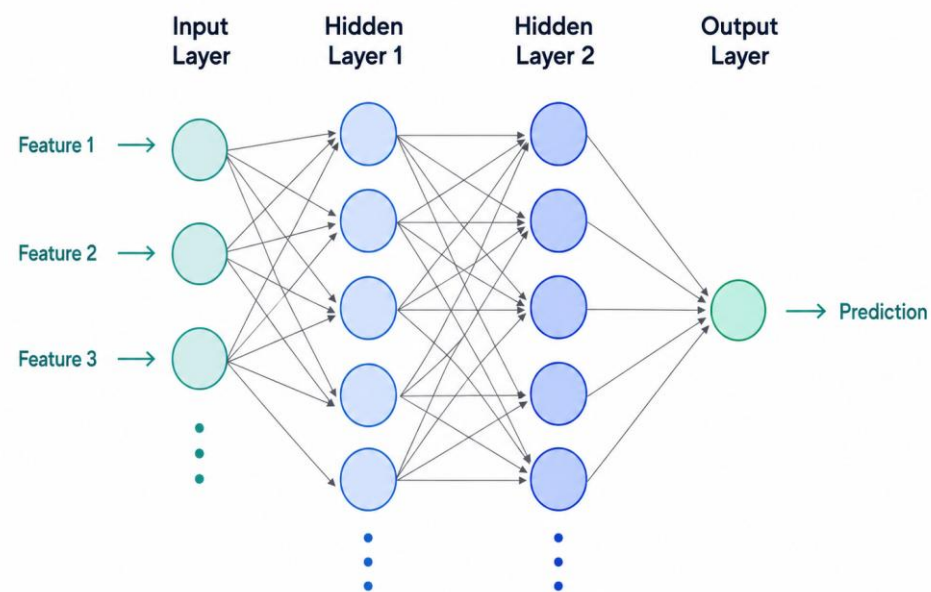


INTRODUCTION TO NEURAL NETWORKS

Neural networks as a key foundation of modern AI

From artificial neurons to deep learning architectures

Conceptual focus: what happens inside the model?





LEARNING OBJECTIVES

- Explain what a neural network is at a conceptual level
 - Identify the main components: neurons, layers, weights, biases, and activation functions
 - Describe how information flows through a neural network
 - Compare common neural network architectures
 - Read a simple Keras neural network structure conceptually
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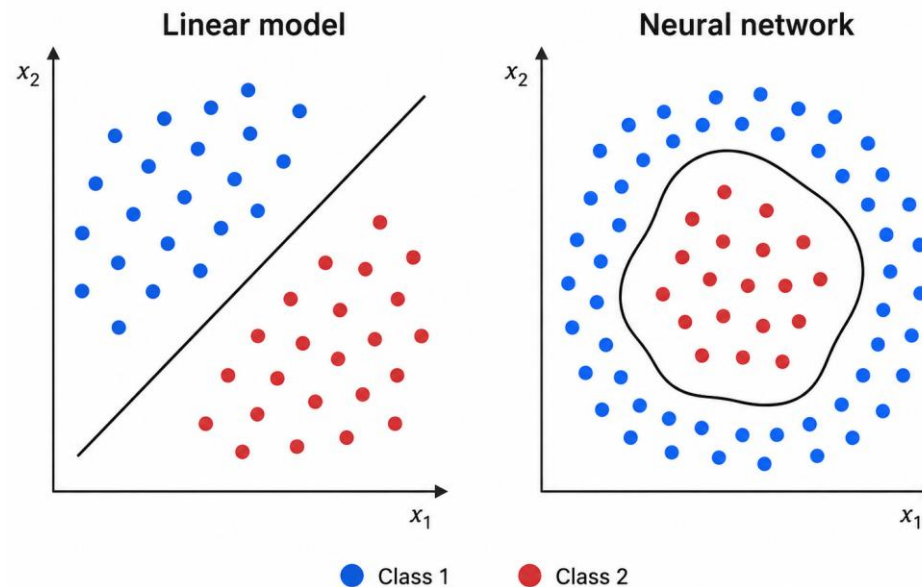
WHY DO WE NEED NEURAL NETWORKS?

Classic machine learning models work well when:

- Patterns are relatively simple
- Features are manually prepared
- Relationships are mostly linear or structured

Neural networks help when:

- Patterns are complex and nonlinear
- Data has many features
- The model needs to learn useful internal representations
- Tasks involve images, text, speech, sequences, or large-scale prediction

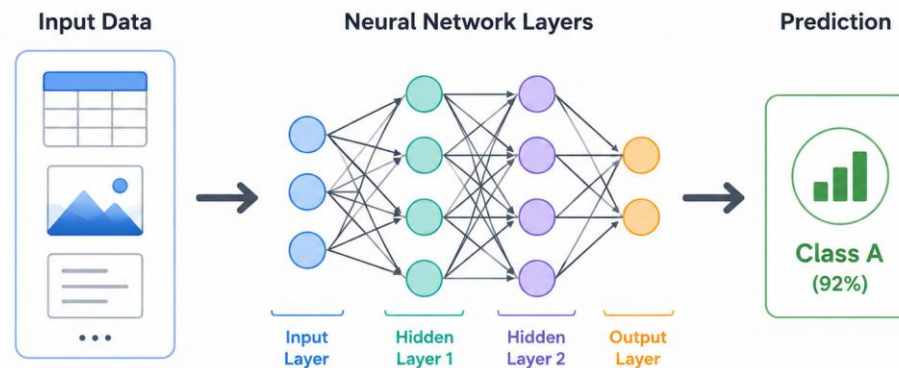




WHAT IS A NEURAL NETWORK?

A neural network is a computational model that:

- Receives input data
- Transforms it through layers of artificial neurons
- Learns weights and biases from examples
- Produces predictions or decisions

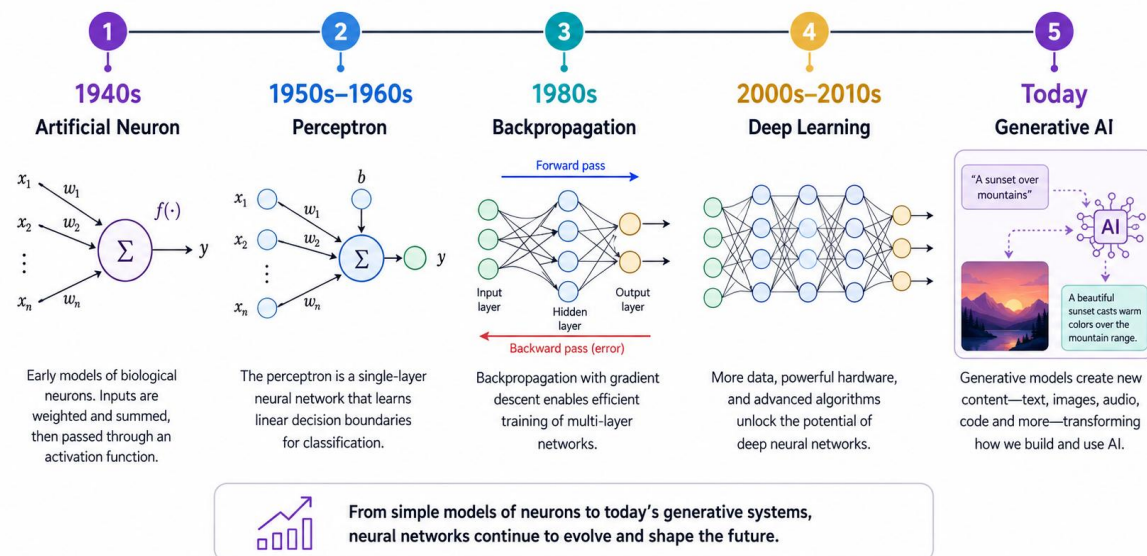




BRIEF HISTORY OF NEURAL NETWORKS

- **1940s–1950s:** Early artificial neuron models
McCulloch-Pitts neuron and Hebbian learning
- **1950s–1960s:** The perceptron
Early model for binary classification
- **1980s:** Backpropagation gains attention
More practical training of multilayer networks
- **2000s–2010s:** Deep learning era
More data, better hardware, better algorithms
- **Today:** Foundation models, generative AI, multimodal AI, and AI agents

A Timeline of Neural Network History

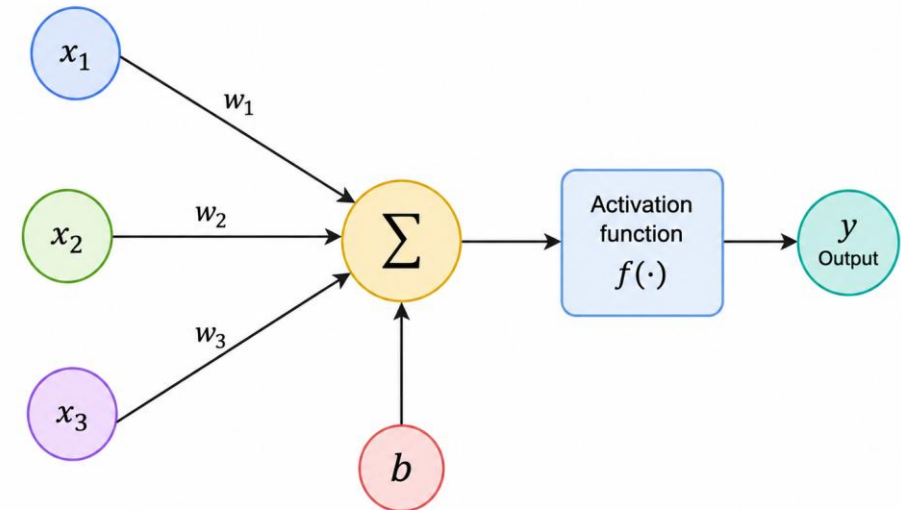




THE ARTIFICIAL NEURON

An artificial neuron:

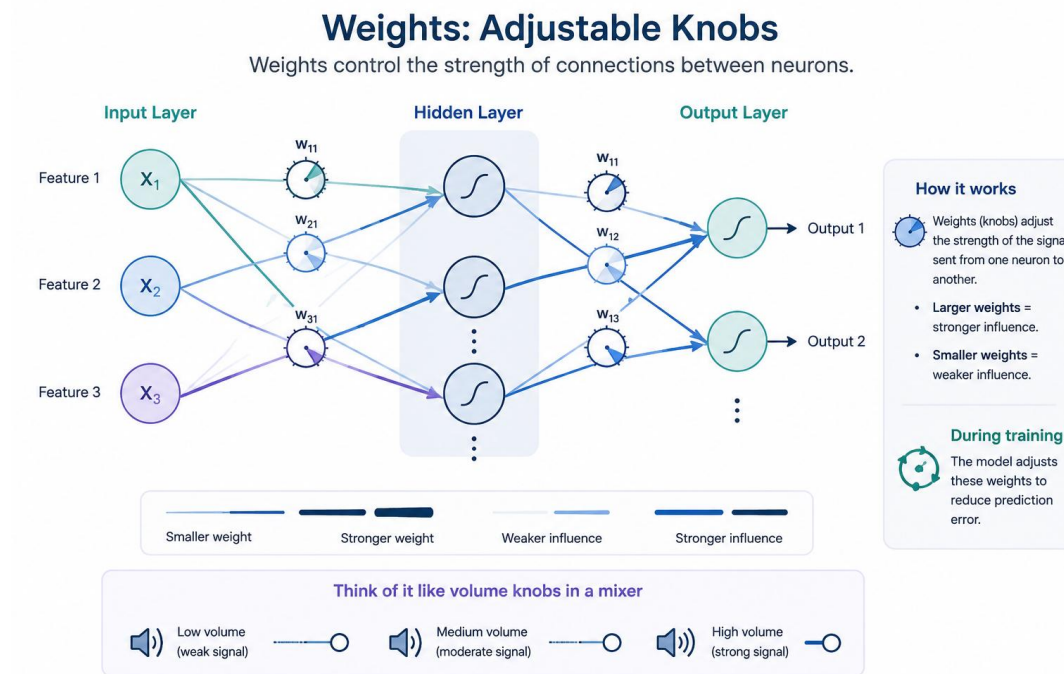
- Receives input values
- Multiplies each input by a weight
- Adds a bias
- Applies an activation function
- Sends the result to the next layer





WEIGHTS AND BIASES

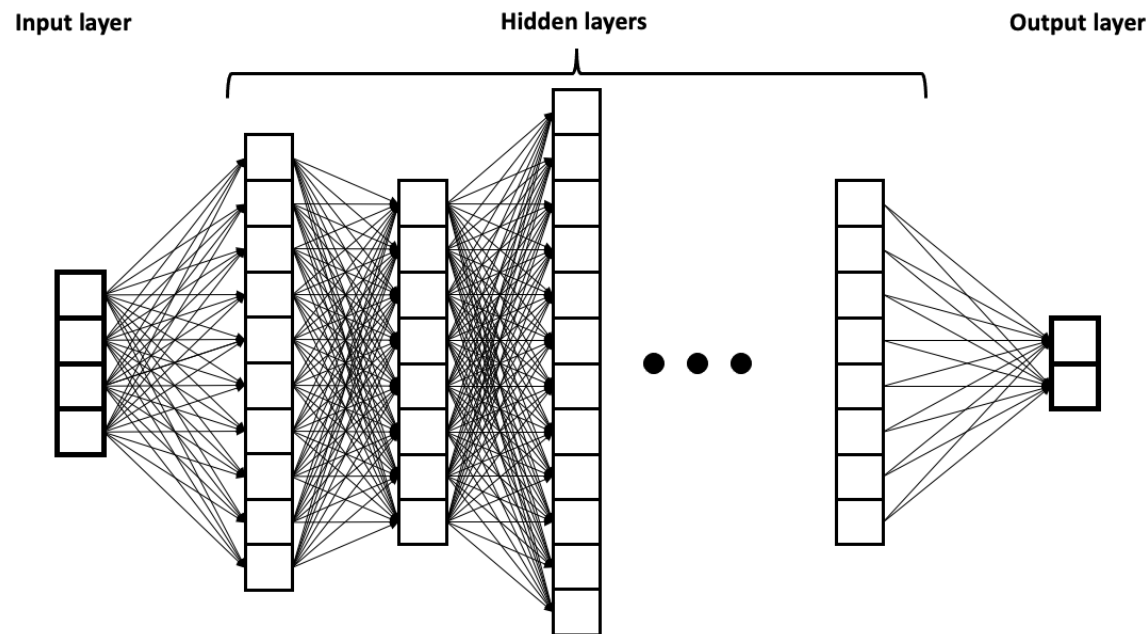
- Weights control the importance of each input
- Biases allow neurons to shift their decision behavior
- During training, the model adjusts weights and biases
- Learning means finding parameter values that reduce prediction error





LAYERS: INPUT, HIDDEN, AND OUTPUT

- **Input layer:** receives features from the dataset
- **Hidden layers:** learn intermediate patterns
- **Output layer:** produces the final prediction

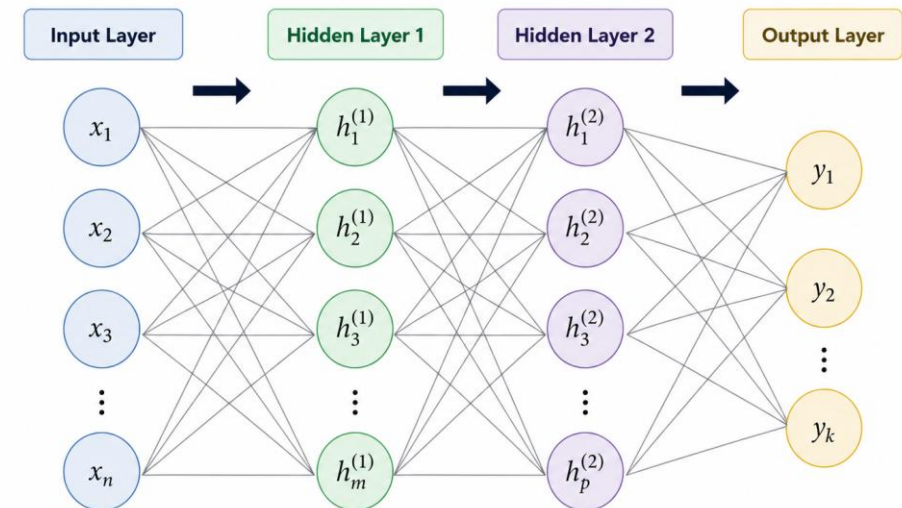




FEEDFORWARD PROCESS

Feedforward is the process where:

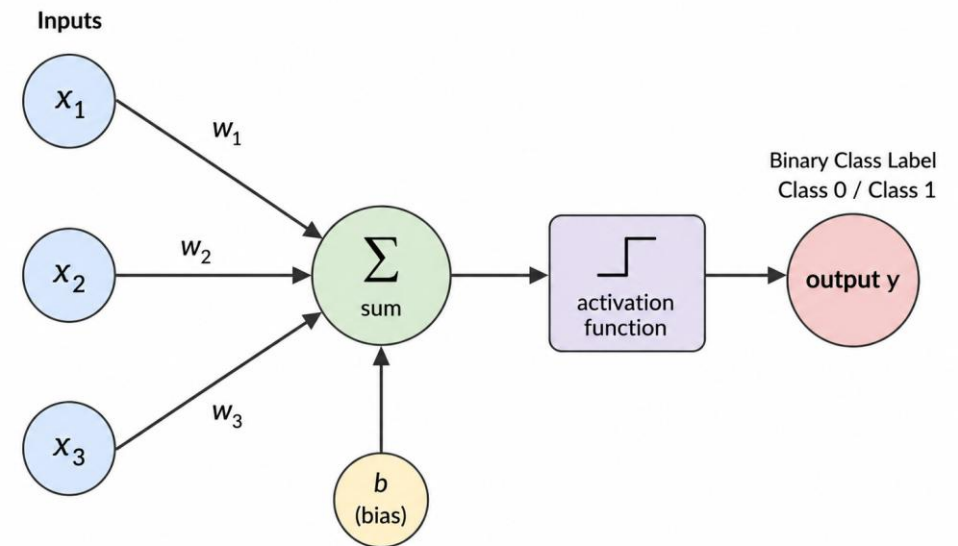
- Inputs enter the network
- Each layer transforms the data
- Outputs from one layer become inputs to the next
- The final layer produces the prediction





THE PERCEPTRON: THE SIMPLEST NEURAL NETWORK

- Takes multiple inputs
- Computes a weighted sum
- Applies an activation rule
- Produces one output



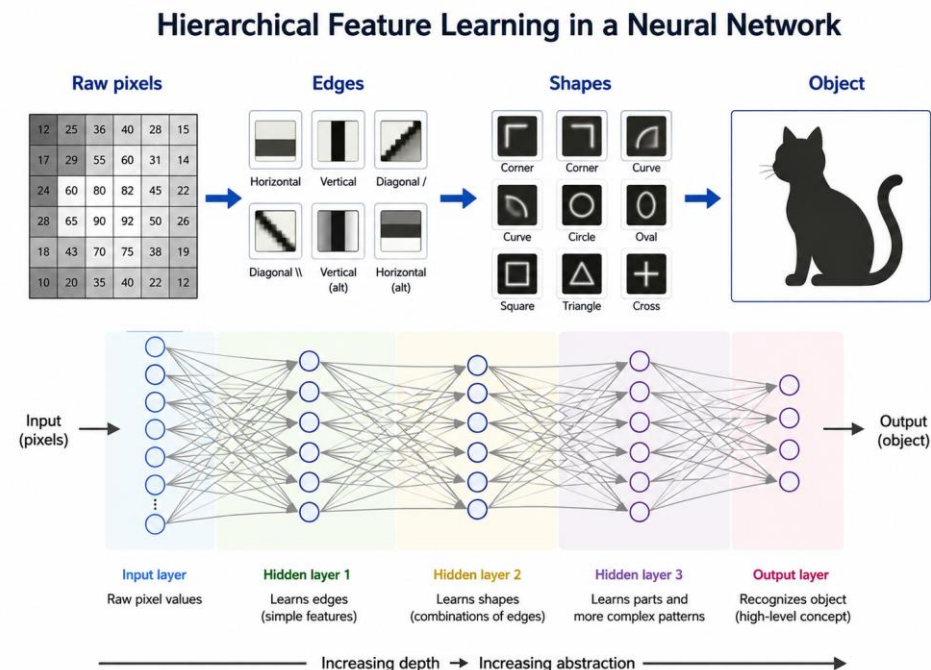


FROM PERCEPTRON TO MULTILAYER NETWORKS

One neuron can learn simple patterns.

Many neurons and layers can learn:

- Feature combinations
- Nonlinear relationships
- Hierarchical representations
- More complex decision boundaries





COMMON NEURAL NETWORK ARCHITECTURES

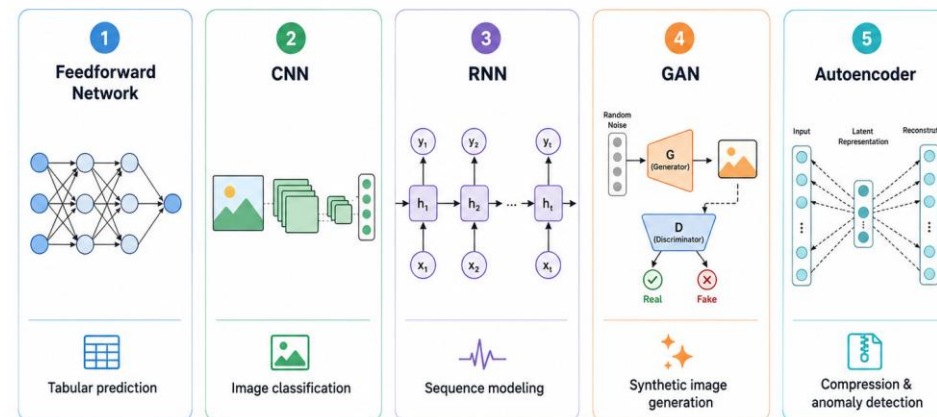
Feedforward networks: general tabular prediction

CNNs: images and spatial patterns

RNNs: sequences and time-dependent data

GANs: generating realistic synthetic data

Autoencoders: compression, reconstruction, anomaly detection





ARCHITECTURE EXAMPLES

Problem	Suitable architecture
Predict customer churn from tabular data	Feedforward network
Classify medical images	CNN
Predict next word in a sentence	RNN or transformer-style model
Detect unusual transactions	Autoencoder
Generate synthetic images	GAN or generative model



CONCEPTUAL READING OF A KERAS MODEL

Example structure from the notebook:

```
model = Sequential()  
model.add(Dense(units=3, input_shape=(3,), activation='relu'))  
model.add(Dense(units=5, activation='relu'))  
model.add(Dense(units=1, activation='sigmoid'))
```

- **Sequential model:** layers are stacked one after another
 - **First dense layer:** receives 3 input features
 - **Hidden layer:** learns 5 intermediate patterns
 - **Output layer:** produces one probability
 - **Sigmoid output:** suitable for binary classification
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KEY TAKEAWAYS

- Neural networks learn complex relationships through layers of neurons
 - Weights and biases are the parameters adjusted during training
 - Hidden layers transform raw features into useful internal representations
 - Different architectures are designed for different data types and tasks
 - A Keras model definition is a compact description of a network architecture
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Technology Elevated